

## ER Diagram - International Airport Database

### 1. System Description

The Airport Database manages airport operations, including passenger bookings, flights, airlines, and booking details. The system tracks airports, the flights that depart from and arrive at them, the airlines that operate those flights, passengers and the bookings they make, any changes made to a booking, boarding passes issued for a booking, baggage registered for a booking, baggage checks performed on registered baggage, and security checks performed on passengers.

### 2. Identified Entities

- Airport
- Airline
- Flight
- Passenger
- Booking
- Booking\_Change
- Boarding\_Pass
- Baggage
- Baggage\_Check
- Security\_Check

### 3. Entities and Attributes

#### 3.1 Airport

| Attribute    | Data type    | Key      |
|--------------|--------------|----------|
| airport_id   | INT          | PK       |
| airport_name | VARCHAR(150) | NOT NULL |
| country      | VARCHAR(100) | NOT NULL |
| state        | VARCHAR(100) | -        |
| city         | VARCHAR(100) | NOT NULL |
| created_at   | TIMESTAMP    | NOT NULL |
| updated_at   | TIMESTAMP    | NOT NULL |

### 3.2 Airline

| Attribute    | Data type    | Key              |
|--------------|--------------|------------------|
| airline_id   | INT          | PK               |
| airline_code | VARCHAR(10)  | UNIQUE, NOT NULL |
| name         | VARCHAR(150) | NOT NULL         |
| country      | VARCHAR(100) | NOT NULL         |
| created_at   | TIMESTAMP    | NOT NULL         |
| updated_at   | TIMESTAMP    | NOT NULL         |

### 3.3 Flight

| Attribute                | Data type   | Key           |
|--------------------------|-------------|---------------|
| flight_id                | INT         | PK            |
| airline_id               | INT         | FK -> Airline |
| departure_airport_id     | INT         | FK -> Airport |
| arrival_airport_id       | INT         | FK -> Airport |
| departing_gate           | VARCHAR(10) | -             |
| arriving_gate            | VARCHAR(10) | -             |
| scheduled_departure_time | DATETIME    | NOT NULL      |
| scheduled_arrival_time   | DATETIME    | NOT NULL      |
| actual_departure_time    | DATETIME    | -             |
| actual_arrival_time      | DATETIME    | -             |
| created_at               | TIMESTAMP   | NOT NULL      |
| updated_at               | TIMESTAMP   | NOT NULL      |

### 3.4 Passenger

| Attribute              | Data type    | Key              |
|------------------------|--------------|------------------|
| passenger_id           | INT          | PK               |
| first_name             | VARCHAR(100) | NOT NULL         |
| last_name              | VARCHAR(100) | NOT NULL         |
| gender                 | VARCHAR(20)  | -                |
| date_of_birth          | DATE         | NOT NULL         |
| country_of_citizenship | VARCHAR(100) | NOT NULL         |
| country_of_residence   | VARCHAR(100) | NOT NULL         |
| passport_number        | VARCHAR(30)  | UNIQUE, NOT NULL |
| created_at             | TIMESTAMP    | NOT NULL         |
| updated_at             | TIMESTAMP    | NOT NULL         |

### 3.5 Booking

| Attribute        | Data type     | Key             |
|------------------|---------------|-----------------|
| booking_id       | INT           | PK              |
| flight_id        | INT           | FK -> Flight    |
| passenger_id     | INT           | FK -> Passenger |
| status           | VARCHAR(30)   | NOT NULL        |
| booking_platform | VARCHAR(50)   | NOT NULL        |
| ticket_price     | DECIMAL(10,2) | NOT NULL        |
| created_at       | TIMESTAMP     | NOT NULL        |
| updated_at       | TIMESTAMP     | NOT NULL        |

### 3.6 Booking\_Change

| Attribute          | Data type    | Key           |
|--------------------|--------------|---------------|
| change_id          | INT          | PK            |
| booking_id         | INT          | FK -> Booking |
| change_description | VARCHAR(255) | NOT NULL      |
| created_at         | TIMESTAMP    | NOT NULL      |
| updated_at         | TIMESTAMP    | NOT NULL      |

### 3.7 Boarding\_Pass

| Attribute        | Data type  | Key           |
|------------------|------------|---------------|
| boarding_pass_id | INT        | PK            |
| booking_id       | INT        | FK -> Booking |
| seat             | VARCHAR(5) | NOT NULL      |
| boarding_time    | DATETIME   | NOT NULL      |
| created_at       | TIMESTAMP  | NOT NULL      |
| updated_at       | TIMESTAMP  | NOT NULL      |

### 3.8 Baggage

| Attribute  | Data type    | Key           |
|------------|--------------|---------------|
| baggage_id | INT          | PK            |
| booking_id | INT          | FK -> Booking |
| weight_kg  | DECIMAL(5,2) | NOT NULL      |
| created_at | TIMESTAMP    | NOT NULL      |
| updated_at | TIMESTAMP    | NOT NULL      |

### 3.9 Baggage\_Check

| Attribute        | Data type   | Key             |
|------------------|-------------|-----------------|
| baggage_check_id | INT         | PK              |
| booking_id       | INT         | FK -> Booking   |
| passenger_id     | INT         | FK -> Passenger |
| check_result     | VARCHAR(30) | NOT NULL        |
| created_at       | TIMESTAMP   | NOT NULL        |
| updated_at       | TIMESTAMP   | NOT NULL        |

### 3.10 Security\_Check

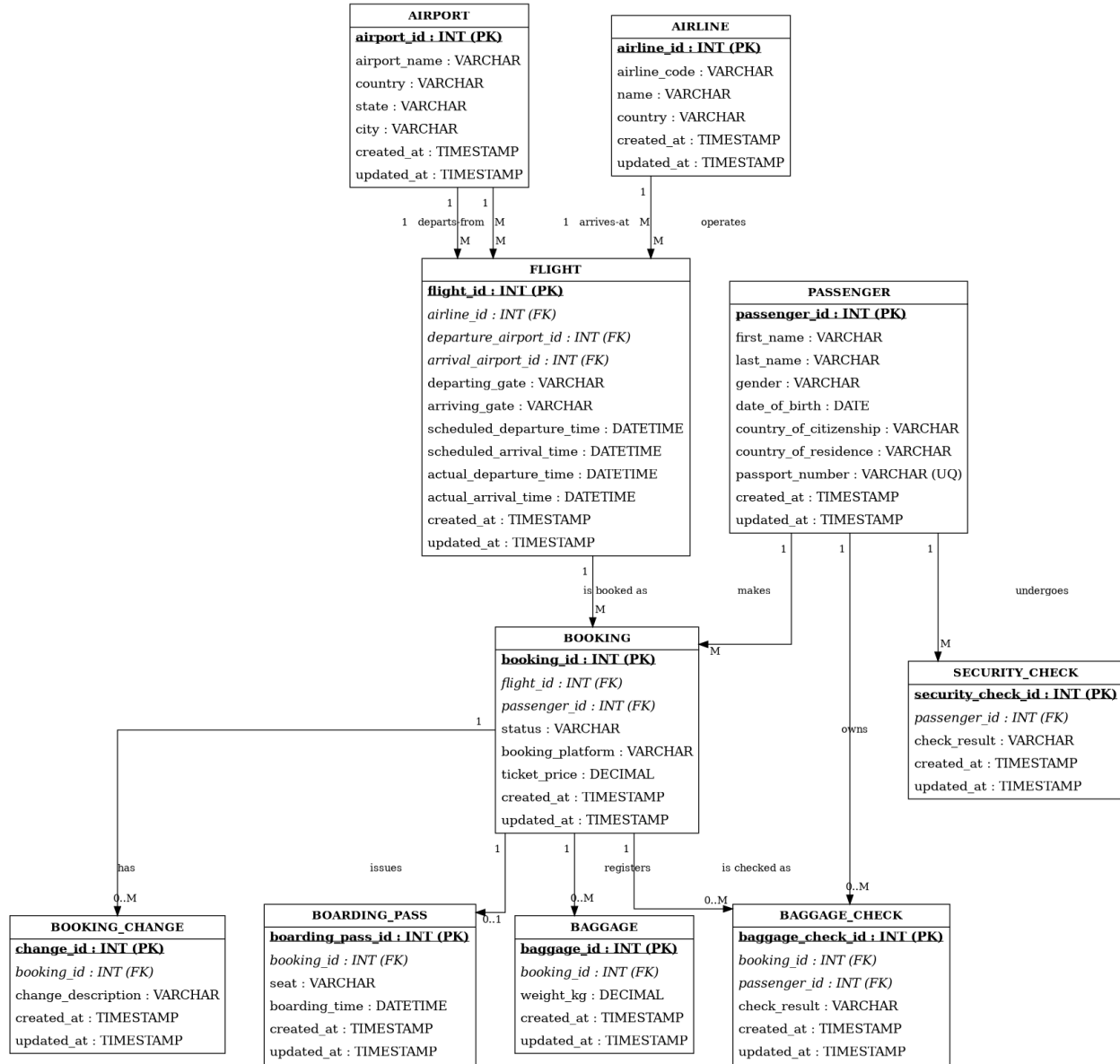
| Attribute         | Data type   | Key             |
|-------------------|-------------|-----------------|
| security_check_id | INT         | PK              |
| passenger_id      | INT         | FK -> Passenger |
| check_result      | VARCHAR(30) | NOT NULL        |
| created_at        | TIMESTAMP   | NOT NULL        |
| updated_at        | TIMESTAMP   | NOT NULL        |

#### 4. Identified Relationships

| #  | Entities                     | Cardinality | Participation                              | Description                                                               |
|----|------------------------------|-------------|--------------------------------------------|---------------------------------------------------------------------------|
| 1  | Airport - Flight (departure) | 1 : M       | Airport optional, Flight mandatory         | One airport is the departure point of many flights                        |
| 2  | Airport - Flight (arrival)   | 1 : M       | Airport optional, Flight mandatory         | One airport is the arrival point of many flights                          |
| 3  | Airline - Flight             | 1 : M       | Airline optional, Flight mandatory         | One airline operates many flights                                         |
| 4  | Flight - Booking             | 1 : M       | Flight optional, Booking mandatory         | One flight can have many bookings                                         |
| 5  | Passenger - Booking          | 1 : M       | Passenger optional, Booking mandatory      | One passenger can make many bookings                                      |
| 6  | Booking - Booking_Change     | 1 : 0..M    | Booking optional (0 or more changes)       | A booking may have zero or more recorded changes                          |
| 7  | Booking - Boarding_Pass      | 1 : 0..1    | Booking optional                           | A booking results in at most one boarding pass                            |
| 8  | Booking - Baggage            | 1 : 0..M    | Booking optional (baggage is not required) | A booking may include zero or more registered baggage items               |
| 9  | Booking - Baggage_Check      | 1 : 0..M    | Booking optional                           | Baggage belonging to a booking is checked; results are logged per booking |
| 10 | Passenger - Baggage_Check    | 1 : 0..M    | Passenger optional                         | A passenger is linked to the baggage check records of their baggage       |
| 11 | Passenger - Security_Check   | 1 : M       | Passenger mandatory                        | Each passenger undergoes one or more security checks                      |

## 5. ER Diagram

Entity-Relationship Diagram - International Airport Database



## 6. Legend

- Each box represents an entity (a table in the database). The top row shows the entity name.
- Bold, underlined attribute = Primary Key (PK) - uniquely identifies each row of the entity.
- Italic attribute = Foreign Key (FK) - references the primary key of another entity, implementing the relationship.
- Plain attribute = regular column (UQ = unique constraint, NOT NULL where required by the system description).
- Lines between entities represent relationships; the label on the line names the relationship.
- Numbers next to the line ends show cardinality: 1 = exactly one, M = many, 0..1 = zero or one (optional), 0..M = zero or many (optional).
- created\_at / updated\_at columns appear in every entity and record when a row was created and last modified, as required by the system description.

*Note: Baggage\_Check stores both booking\_id and passenger\_id, following the checking process described in the system specification (baggage belonging to a booking is checked, and the check result is also associated with the passenger who owns that baggage).*